

NAJATH AKRAM, PH.D.

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SUMMARY

Signal processing engineer, ten years in wireless, the last six on LTE and 5G NR O-RAN radio units. At Airspan, owns the bit-accurate DFE and low-PHY reference models firmware is signed off against, and the lab team's fronthaul test tools; at Jabil, wrote the DFE architecture and 3GPP implementation specifications the digital design team built to and was promoted twice in four years to Principal Engineer. Ph.D. dissertation in reduced-hardware beamforming receivers; 10+ IEEE publications.

CORE SKILLS

Wireless systems: 5G NR and LTE physical layer; O-RAN fronthaul (split 7.2x, WG4 CUS-plane, eCPRI, BFP compression); 3GPP TS 38.211, 38.104, 38.141 and 36.141; multiband TDD and FDD radio unit architecture; EVM, ACLR, OBUE and PIM analysis

DSP and modeling: DFE design (DUC and DDC, NCO, channel and pulse-shaping filters, interpolation and decimation chains); CFR and cancellation-pulse design; PRACH detection; fixed-point and bit-exact modeling against vendor IP C-models; BFP codecs

Platforms and tools: MATLAB (5G, LTE and HDL toolboxes, App Designer, Compiler); AMD/Xilinx RFSoc and DFE IP; Python; Keysight 89600 VSA

AI engineering: Anthropic Claude API and Agent SDK; multi-agent pipelines with schema-validated outputs; per-call cost ledgers and budgets; evaluation harnesses; TypeScript, Next.js, PostgreSQL

EXPERIENCE

AIRSPAN NETWORKS

Houston, Texas

Signal Processing Engineer

January 2025 to present

O-RAN 7.2x radio units: bit-accurate DFE and low-PHY reference modeling, FPGA IP verification, and fronthaul test tooling.

- Developed and maintain the bit-accurate DL/UL DFE + low-PHY reference models that generate the RTL golden vectors, so RTL is compared bit for bit rather than to a tolerance: NR/LTE test models through IFFT/FFT, CFR, the Xilinx channel-filter/mixer IP C-models, and O-RAN BFP at 8/9/10/12/14-bit, across 3-100 MHz including combined-band carriers (e.g. B28+B20).
- Built the bit-accurate PRACH receive-chain model (Xilinx PRACH DDC and decimation, CP removal, FFT, subcarrier extraction, BFP and U-plane packing, detection) covering every LTE format and all NR FR1 formats, L_{RA} 839 and 139, FDD and TDD.
- Built an O-RAN fronthaul power and PAPR tool: per-eAxC dBFS power, grid-domain PAR, per-tone PAR, and oversampled time-domain PAPR reported separately, auto-detecting BFP width and eAxC bit split and deriving the compressed-domain full-scale reference per O-RAN WG4 CUS 8.1.3.1. Shipped as MATLAB Compiler GUI and CLI executables plus a Python version.
- Built an uplink EVM and PRACH analyzer that demodulates PUSCH and PRACH from 7.2x U-plane captures, with views aligned to the Keysight 89600 VSA, and an RU-to-DU full-scale alignment check that emulates the DU's fixed-point ingest (shift, container width, wrap or saturate) against the per-PRB BFP exponents to separate RU misconfiguration from DU ingest error.
- Built the expected versus captured IQ comparison GUI used in bring-up: cross-correlation delay alignment, overlaid PSD and PSD difference, and dBFS power readout for hex, BFP, mat, and raw binary captures.
- Re-clocked CFR from 491.52 to 245.76 MSPS and redesigned the cancellation pulses and channel filters for 3 to 20 MHz (a 3 MHz channel filter is not realizable at 30.72 MSPS) to meet an operator's 3 MHz carrier requirement, tuning cutoff and pulse length against a per-symbol constellation EVM metric rather than the vendor's RMS figure: 2.17 to 2.23% EVM at 3, 5, and 10 MHz inside ACLR and OBUE margin.
- Built a multiband PIM planner that catalogs IM3, IM5, and IM7 products for multi-carrier downlink and flags uplink-band hits; checked by hand against a B71, B29, and B14 stack.

JABIL INC.

Warren, New Jersey and Houston, Texas

Principal Wireless Systems Design Engineer

December 2023 to August 2024

Lead Wireless Systems Design Engineer

September 2021 to December 2023

Senior FPGA Design Engineer

December 2020 to September 2021

- Led the software and firmware workstreams for LTE and 5G NR O-RAN radio units from single-band to tri-band through commercial release: defined NR implementation requirements, wrote the architecture and implementation documents the digital design team built to, weighed massive MIMO and O-RAN functional-split options against implementation complexity, resources, and timelines, and briefed customers on passive intermodulation.
- Primary author of the DFE System Design Documents for two radio lines, a B3 FDD radio (5 to 40 MHz) and a 100 MHz TDD radio: the interpolation chain to 983.04 MSPS, pulse-shaping filters derived from TS 38.104 guard-band rules and grouped into five classes covering thirteen bandwidths, CFR cancellation-pulse configuration, and uplink decimation and PRACH routing.
- Authored the firmware implementation guides for the NR symbol phase-compensation term of TS 38.211 section 5.4 (quadrant-folded trigonometric LUT, 4097-entry table, 21-bit minimum phase accumulator, numerology-dependent parameters, since a wrong term shows up as a rotated constellation; revised through v1.3) and for RFSoc ZU67DR pre-emphasis (per-subcarrier 16-bit LUTs flattening pulse-shaping passband droop for 5 to 30 MHz, with the LTE and LTE plus NB-IoT edge cases per TS 38.104 section 5.7.3).

- Wrote the uplink DFE receiver test plan for LTE, NR and MSR against TS 36.141 and TS 38.141: reference sensitivity, dynamic range, in-channel and adjacent-channel selectivity, and narrowband, in-band and out-of-band blocking, with bottom, mid and top frequency coverage per band and VRB-offset sweeps.
- Designed 8-block PRACH detection within the Xilinx PRACH core's 4K-FFT limit by cascading two 4K FFTs as a filter bank with per-block phase-rotation correction, and mapped LTE and NR PRACH subcarriers onto the core's FFT sizes and decimation ratios.
- Ran filter design, CFR simulation and cancellation-pulse generation for uplink, downlink and PRACH DFE firmware: a hard-clipper threshold sweep in 0.1 dB steps that set the 7.8 dB PAR operating point for E-TM 3.1, a first- and second-difference slew-rate detector for post-IFFT overflow, and cancellation-pulse analysis delivered to a major OpenRAN program partner in 2022.
- Generated RTL stimulus vectors, integrated Xilinx hard-IP C-models, developed executable models for UVM testbenches, and wrote PCAP tooling to extract O-RAN C-plane and U-plane data for bit-accurate system modeling.

FLORIDA INTERNATIONAL UNIVERSITY

Miami, Florida

Graduate Research Assistant

2018 to 2020

- Developed digital and mixed-domain hardware-reduction algorithms for massive MIMO receivers, the basis of the dissertation and most of the 10+ IEEE publications, all on reduced-ADC array architectures.
- Implemented a 28 GHz digital beamforming array receiver with 800 MHz per channel on Xilinx RFSoc (ZCU111, ZCU1275, ZCU1285), cutting ADC count by 75% with frequency multiplexing, and showed a further 50% ADC reduction for 2D phased arrays over the air.

MATHWORKS

Natick, Massachusetts

Engineering Intern

2019

- Enhanced HDL Verifier workflows and developed Ethernet PHY-interface IP across Artix, Kintex and Virtex FPGAs; two awards in the fall 2019 intern hackathon.

THE UNIVERSITY OF AKRON

Akron, Ohio

Graduate Research Assistant

2016 to 2018

- DARPA, NSF, and AFRL-funded research on multidimensional RF systems: approximate-DFT beamforming cores and multidimensional delta-sigma ADC architectures for phased-array receivers.

SYNOPSYS INC.

Colombo, Sri Lanka

Product Verification Intern

2015

- Automated product-verification test cases in Perl across VHDL and SystemVerilog ASIC flows with Spyglass and Xilinx FPGAs.

INDEPENDENT AI SYSTEMS

- **Salient:** USMLE exam practice with an AI patient that never gives away the diagnosis. All case details are pre-written and checked, so the model cannot invent a result, and a leak guard blocks any answer that might reveal the diagnosis. Six model roles, each sized to its task; a per-call cost ledger with daily budgets; grading aligned with published exam rubrics. 700 unit tests and 24 browser journeys.
- **TaxLens:** Rebuilds a year's small-business books from raw bank and card statements into a locked ledger, financials, and tax worksheets (Schedule C, 1120-S, 1065, 1120). Routes each step to a model sized to its risk in a multi-agent pipeline. Each deduction carries an IRS citation, an evidence rating, and a confidence score. A Section 274(d) guard runs in code on every write; classifications are append-only and hashed on lock. In one real client year, it caught \$10K of double-counted bounced payments that manual review had missed. 400+ unit tests. It does not file.

EDUCATION

FLORIDA INTERNATIONAL UNIVERSITY Ph.D., Electrical and Computer Engineering, 2020

Miami, Florida

UNIVERSITY OF RUHUNA B.Sc. (Hons), Electrical and Information Engineering, 2016

Galle, Sri Lanka

NATIONAL INSTITUTE OF BUSINESS MANAGEMENT A.S., Business Management, 2015

Galle, Sri Lanka

Affiliations: IEEE, member since 2013; Tau Beta Pi Engineering Honor Society, 2019 to present

SELECTED PUBLICATIONS

- Frequency-Multiplexed Array Digitization for MIMO Receivers: 4-Antennas/ADC at 28 GHz on Xilinx ZCU-1285 RF SoC, IEEE Access, vol. 9, 2021.
- Spacetime Frequency-Multiplexed Digital-RF Array Receivers with Reduced ADC Count, IEEE Transactions on Circuits and Systems II: Express Briefs, vol. 68, 2021.
- Massive-MIMO and Digital mm-Wave Arrays on RF-SoCs using FDM for M-Fold Increase in Antennas per ADC/DAC, 2021 IEEE Space Hardware and Radio Conference (SHaRC), 2021.
- Fast Radix-32 Approximate DFTs for 1024-Beam Digital RF Beamforming, IEEE Access, vol. 8, 2020.
- A Direct-Conversion Digital Beamforming Array Receiver with 800 MHz Channel Bandwidth at 28 GHz using Xilinx RF SoC, IEEE COMCAS, Israel, 2019.
- Multiport ADCs for Microwave Focal Plane Array Dish Receivers, IEEE International Symposium on Circuits and Systems, Italy, 2018.

10+ IEEE publications, an AFRL technical report, and a dissertation; full list at najathakram.info.